

Mathematics 1

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The derivative of a function $y(x)$ is defined as

$$\frac{d}{dx}y(x) = \lim_{\delta x \rightarrow 0} \frac{y(x + \delta x) - y(x)}{(x + \delta x) - x} = y'$$

This derivative is obtained in a limit process according to

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

As an example consider a straight line with the function

$$y(x) = m \cdot x + b$$

m : slope b : determines the point of intersection with the axis

Betrachte $y(x) = 3 \cdot x - 5$

table

x	y
-3	-14
-1	-8
0	-5
1	-2
3	4
5	10
7	16

differences

Δx	Δx	Δy	Δy
-3-(-1)	-2	-14-(-8)	-6
-3-0	-3	-14-(-5)	-9
0-7	-7	-5-16	-21
1-5	-4	-2-10	-12
3-(-3)	6	4-(-14)	18
1-(-1)	2	-2-(-8)	6
7-(-3)	10	16-(-14)	30

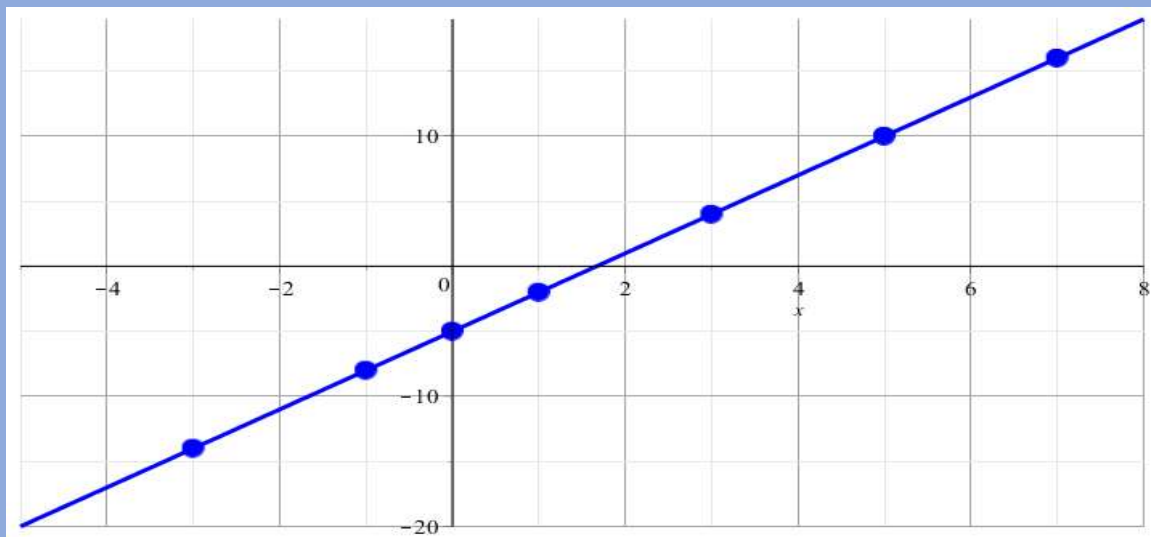
calculate $\frac{\Delta y}{\Delta x}$

$\frac{\Delta y}{\Delta x}$
3
3
3
3
3
3
3

Slope of
the
straight
line

$$y = f(x) = 3 \cdot x - 5$$

graph



Consider $y(x) = 3 \cdot x - 5$

Remark: For a straight line the slope is the same for all values of x .

The limit

$$\begin{aligned}\frac{d}{dx}y(x) &= \lim_{\delta x \rightarrow 0} \frac{y(x + \delta x) - y(x)}{(x + \delta x) - x} = \lim_{\delta x \rightarrow 0} \frac{(3(x + \delta x) - 5) - (3x - 5)}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{3x - 5 + 3\delta x - 3x + 5}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{3 \cdot \delta x}{\delta x} = \lim_{\delta x \rightarrow 0} 3 = 3\end{aligned}$$

demonstrates, that the derivative is equivalent to the value of the slope.
A straight line has a constant slope for all values of x .

In general: for a straight line with $y(x) = m \cdot x + b$ the derivative will yield the slope m , or

$$\frac{d}{dx}y(x) = y' = m$$

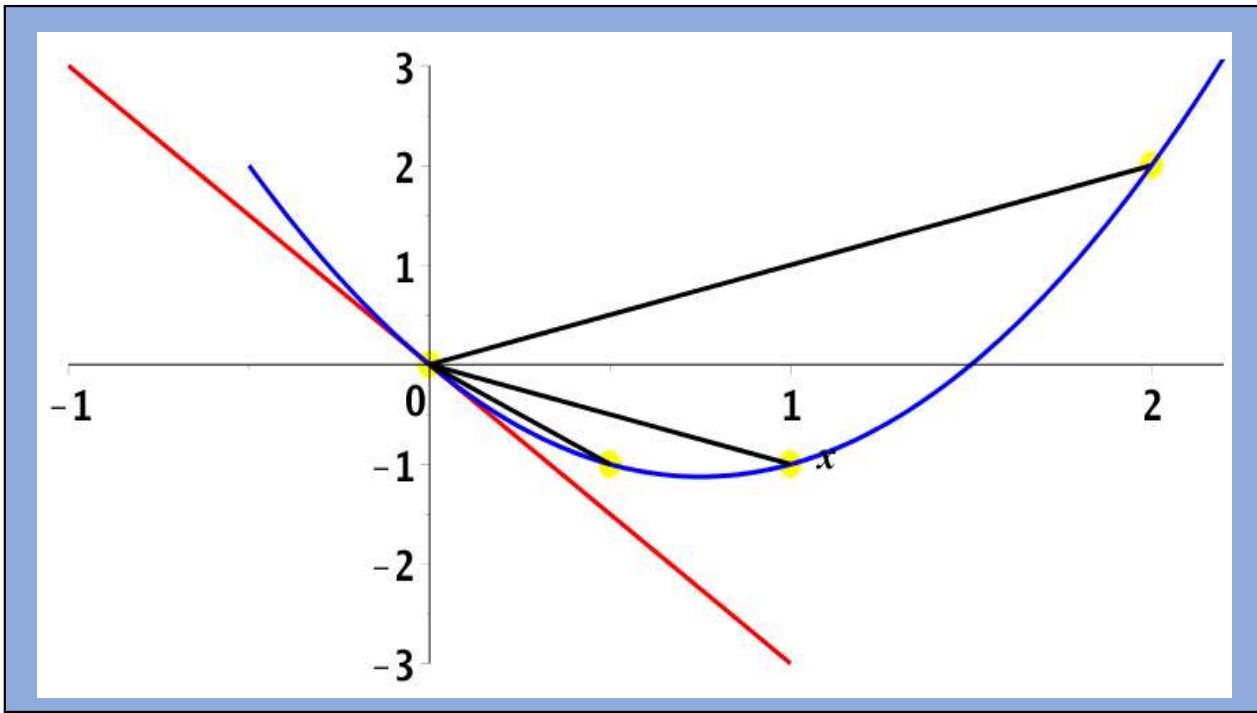
Die difference ratio yields the derivative as this ratio does not depend on Δx .

Question: How can be treat a function which is not a straight line, such as for example $y(x) = a \cdot x^2 + b \cdot x$?

Calculations for $y(x) = 2 \cdot x^2 - 3 \cdot x$

Choose $\delta x > 0$ and keep the left point fixed at $x_0 = 0$.

x_1	x_2	y_1	y_2	$\frac{\Delta y}{\Delta x}$
0	2	0	2	1
0	1.5	0	0	0
0	1	0	-1	-1
0	0.5	0	-1	-2
0	0.1	0	-0.28	-2.8
0	0.02	0	-0.0592	-2.96
0	0.004	0	-0.011968	-2.992
0	0.0008	0	-0.0023987	-2.9984
0	0.00016	0	-0.0004799	-2.99968
0	0.000032	0	-9.6E-05	-2.999936
0	0.0000064	0	-1.92E-05	-2.9999872
0	0.00000128	0	-3.84E-06	-2.9999974



Calculation for $y(x) = 2 \cdot x^2 - 3 \cdot x$

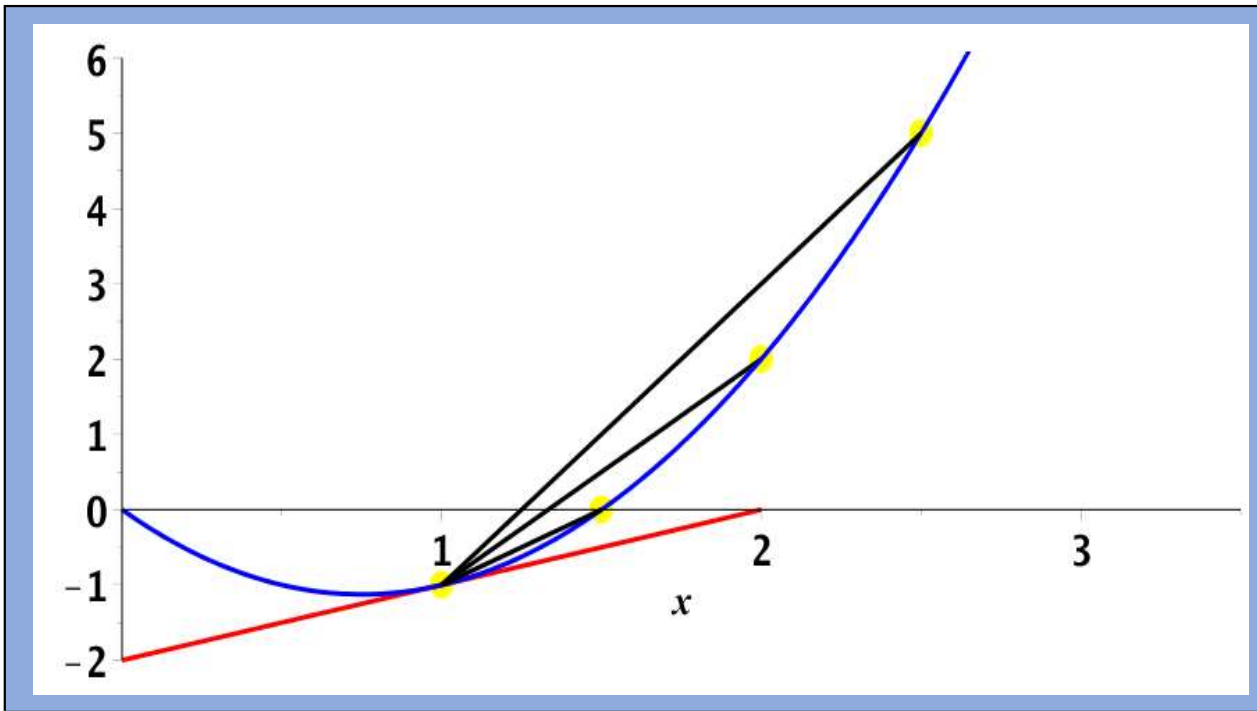
Choose a different point, for example $x_0 = 1$.

x_1	x_2	y_1	y_2	$\frac{\Delta y}{\Delta x}$
1	3	-1	9	5
1	2.5	-1	5	4
1	2	-1	2	3
1	1.5	-1	0	2
1	1.1	-1	-0.88	1.2
1	1.02	-1	-0.9792	1.04
1	1.004	-1	-0.995968	1.008
1	1.0008	-1	-0.9991987	1.0016
1	1.00016	-1	-0.9998399	1.00032
1	1.000032	-1	-0.999968	1.000064
1	1.0000064	-1	-0.9999936	1.0000128
1	1.00000128	-1	-0.9999987	1.00000256

At $x=1$ the value of the limit and the derivative are identical:

$$\begin{aligned} & \frac{d}{dx} y \\ &= \frac{d}{dx} (2 \cdot x^2 - 3 \cdot x) \\ &= 4x - 3 \end{aligned}$$

$$y'(1) = 4 \cdot 1 - 3 = 1$$



Geometric Interpretation of the Derivative

The derivative of a function of x , $y(x)$, namely $\frac{d}{dx}y(x)$, is to be evaluated at the point x_0 . The result is the **slope of the tangent curve at $(x_0, y(x_0))$** .

Notation $\left. \frac{d}{dx}y(x) \right|_{x=x_0}$ (read: derivative evaluated at point x_0)

It is possible to identify (local) maxima and minima, as the tangent straight line has a slope of 0 at these extremal points (the tangent is parallel to the x-axis).

Consider $y(x) = x^3 - 6x^2 + 9x - 7$

The derivative is evaluated as

$$\frac{d}{dx}y(x) = 3x^2 - 12x + 9$$

Maxima and minima are determined by setting the derivative to 0. this requires solving the polynomial of 2nd order, namely $3x^2 - 12x + 9 = 0$

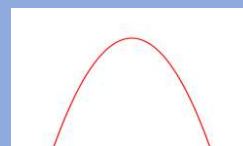
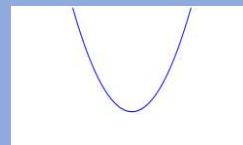
The solutions of $x^2 - 4x + 3 = 0$ are:

$$x_1 = 3 \quad \text{and} \quad x_2 = 1$$

Consider the graph to identify the minimum and the maximum nature of the points.

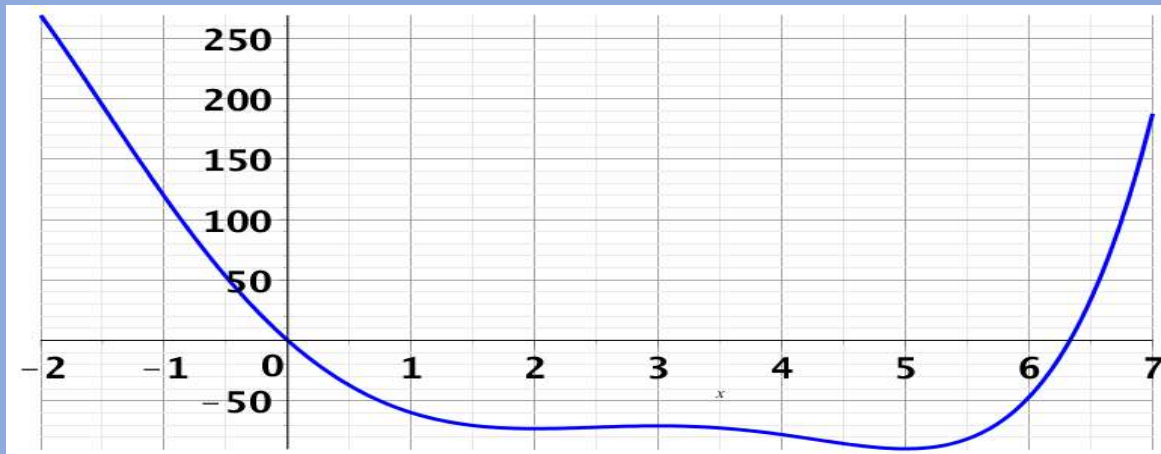
Mathematical Characteristics

	maximum	minimum
Left of extremal point	slope of tangent is positiv	slope of tangent is negativ
right of extremal	slope of tangent is negativ	slope of tangent is positiv
Derivative of tangent slope function	< 0	> 0



Example

Given the function $f(x) = \frac{1}{5}x^5 - \frac{7}{4}x^4 + \frac{1}{3}x^3 + \frac{63}{2}x^2 - 90x$



Determine minima and maxima of this function

Have $f(x) = \frac{1}{5}x^5 - \frac{7}{4}x^4 + \frac{1}{3}x^3 + \frac{63}{2}x^2 - 90x$

derivative: $f' = x^4 - 7x^3 + x^2 + 63x - 90$

In order to determine minima and maxima solve

$$f'(x) = x^4 - 7x^3 + x^2 + 63x - 90 = 0$$

Use trial and error (or any other method) to determine the zero points.

For example try: $x = 2$ and $x = 3$

Check:

$$f'(x = 2) = 2^4 - 7 \cdot 2^3 + 2^2 + 63 \cdot 2 - 90 = 16 - 56 + 4 + 126 - 90 = 0$$

$$f'(x = 3) = 3^4 - 7 \cdot 3^3 + 3^2 + 63 \cdot 3 - 90 = 81 - 189 + 9 + 189 - 90 = 0$$

Simplification of polynomial by Polynomial Division

$$(x^4 - 7x^3 + x^2 + 63x - 90) : (x - 2) = x^3 - 5x^2 - 9x + 45$$

$$\begin{array}{r} x^4 - 2x^3 \\ \hline 0 - 5x^3 \\ -5x^3 + 10x^2 \\ \hline 0 - 9x^2 \\ -9x^2 + 18x \\ \hline 0 + 45x \\ +45x - 90 \\ \hline 0 \end{array}$$

Simplification of polynomial by Polynomial Division

$$(x^3 - 5x^2 - 9x + 45):(x - 3) = x^2 - 2x - 15$$

$$\begin{array}{r} x^3 - 3x^2 \\ \hline 0 - 2x^2 \\ \quad -2x^2 + 6x \\ \hline \quad 0 - 15x \\ \quad \quad -15x + 45 \\ \hline \quad \quad 0 \end{array}$$

Of course we have:

$$(x - 2) \cdot (x - 3) = x^2 - 2x - 3x + 6 = x^2 - 5x + 6$$

$$(x^4 - 7x^3 + x^2 + 63x - 90):(x^2 - 5x + 6) = x^2 - 2x - 15$$

This may be verified by explicit calculation

Solve the equation $x^2 - 2x - 15 = 0$ by explicit calculation

Obtain the zeros as $x = -3$ and $x = 5$.

Therefore the function $f'(x) = x^4 - 7x^3 + x^2 + 63x - 90$ has zeros at $x = -3, 2, 3, 5$.

$$f'(x) = (x + 3) \cdot (x - 2) \cdot (x - 3) \cdot (x - 5)$$

Calculate $f''(x) = 4x^3 - 21x^2 + 2x + 63$ and obtain

$$f''(x = -3) = -240 < 0 \text{ maximum}$$

$$f''(x = 2) = 15 > 0 \text{ minimum}$$

$$f''(x = 3) = -12 < 0 \text{ maximum}$$

$$f''(x = 5) = 48 > 0 \text{ minimum}$$

